ZS50 Camera and Radar Integration Inquiry

Technical integration information and exclusive strategic MOU

We are evaluating the ZS50 camera and associated radar linkage for integration with our existing EPAIL Intelligence operations platform. Please provide the technical and commercial information below so that we can confirm the required hardware, software interfaces, deployment design and support model before procurement.

1 Our software environment

EPAIL Intelligence is a web-based operations platform with a React frontend and a Bun/Hono backend exposing HTTP APIs. It uses PostgreSQL for application data and includes map-based operations views, alert workflows and connected-device integrations. The intended integration is for authorized operators to view live camera feeds, see radar detections and camera tracking status, and access relevant target and device information in EPAIL Intelligence.

Please identify which functions can be accessed directly by a third-party system, which require your supplied Radar and Video Fusion System software, and which depend on a local controller or gateway. We are open to an on-premises or hybrid design, subject to the connectivity, security and licensing details below.

2 Hardware, connectivity and deployment

- **Connection and cabling:** For each camera, radar and controller, list physical interfaces (Ethernet, fibre, serial RS-232/RS-485, discrete I/O or other), connector type, supported cable lengths, and whether Wi-Fi or another wireless link is available. Identify which connections are required for video, PTZ control, radar data, time synchronisation and power.
- **Network and internet:** Can the system operate entirely on a private local network without internet access or vendor cloud services? If internet or cloud access is required or optional, specify why, what data leaves the site, destinations, bandwidth, ports and firewall rules. Explain remote access options, including VPN support.
- **Computer or laptop requirement:** Can the camera and radar operate independently on the network, or must they be connected to a laptop or computer running your software? If a computer is required, does it need to remain switched on for live video, radar detection and tracking? Can EPAIL Intelligence access the equipment directly, or must your desktop software remain open? Please show the simplest supported setup.
- **Power and installation:** Provide voltage/current and power consumption for each component, PoE support (if any), environmental ratings, mounting and grounding requirements, surge/lightning protection, and recommended installation diagram.
- **Network plan:** Provide a reference topology showing radar, camera, OEM fusion/controller software, EPAIL Intelligence server and operator browsers, including IP addressing, ports, protocols, bandwidth and any required gateway or video management system.

3 Video, control and radar data interfaces

- **Video streams:** Confirm supported codecs, resolutions, frame rates and stream counts for visible and thermal channels. Do you provide RTSP/RTP streams, ONVIF Profile T or another ONVIF profile, or a proprietary stream? Share sample stream URLs, authentication method and a test account or simulator. Can the stream be consumed by an external client without opening your desktop application?
- **Camera control:** Document supported PTZ, zoom, focus, preset, tracking start/stop, stabilization and status/telemetry commands. Identify whether these use ONVIF, a documented HTTP/HTTPS API, a vendor SDK, or a proprietary protocol.
- **Radar and target data:** Explain how radar detections and tracks are exposed to external software: documented REST/HTTP API, WebSocket, TCP/UDP messages, serial protocol, SDK, database/event feed, or another method. Provide the schema and example messages, including target ID, latitude/longitude, altitude, range, bearing, speed, classification, confidence, timestamp, track state and camera cueing/status where available.
- **Radar-to-camera linkage:** Describe how the radar cues the camera and how the camera reports acquisition, tracking loss and reacquisition. Is this linkage performed inside the supplied controller, directly between radar and camera, or through an API that EPAIL Intelligence can invoke? Can an external system receive the linked track and issue authorized commands?
- **Time and coordinates:** Specify coordinate reference system and datum, angle conventions, units, time format, synchronisation method (NTP/PTP/GNSS), expected latency and how sensor/camera alignment and calibration are performed.

4 Radar ground coverage and detection performance

Please confirm whether the offered radar detects targets at ground level as well as aerial targets. Please clarify whether ground coverage means detection of people/vehicles on the surface, coverage of terrain at low elevation, or both, and provide performance evidence for the intended use case.

- Provide radar frequency band, instrumented range, azimuth/elevation coverage, minimum detectable target height, minimum range, scan/update rate, and any terrain or horizon constraints.
- Provide detection ranges and probabilities by target class (person, vehicle, drone/bird), target size, speed and aspect, with false-alarm rate and test conditions.
- Explain ground clutter handling, vegetation/building masking, terrain shadowing, rain/wind effects, line-of-sight limits, low-altitude blind zones and radar siting/height requirements.
- Confirm whether radar target coordinates can be output to EPAIL Intelligence in a map-ready reference system and whether ground targets can cue and be tracked by the camera.

5 SDK, API and integration package

For our engineering team, please provide both the documented interface specification and the appropriate integration tools. An API defines the commands and data formats that software exchanges; an SDK is a vendor toolkit, libraries and sample code that help developers use those interfaces. If the OEM fusion software is the only supported integration point, please provide its supported API/SDK and licensing terms.

- English-language user manuals, installation and configuration guides, API/SDK reference, protocol specification, data dictionary, network/security guide and firmware upgrade/rollback procedure.
- English-language software interfaces and operator documentation, including all user-facing labels, alarms, logs and configuration menus. Confirm localization responsibilities for future updates.
- SDK/API packages, supported operating systems and language bindings, version compatibility, code examples, simulator or replay dataset, test credentials, and developer support contact.
- Authentication and authorization model, encryption in transit, certificate support, user/role management, audit logs, vulnerability/patch process and software bill of materials where available.
- Integration acceptance criteria, test plan, OEM engineering support, training, warranty, maintenance, update cadence and end-of-life policy.

6 Requested integration workflow

Please propose a supported workflow in which EPAIL Intelligence can: (a) receive radar detections and track updates; (b) display authorized visible/thermal video; (c) request camera cueing or tracking when permitted; (d) receive camera orientation, zoom/focus and tracking status; and (e) record operator actions and relevant target events. Please distinguish real-time streaming from event/metadata exchange, identify any gateway or media server required, and provide a demonstration or proof-of-concept plan with measurable acceptance criteria.

7 Exclusive strategic MOU

We would also like to enter into an exclusive Memorandum of Understanding with the OEM for the agreed product and market scope. Please confirm your willingness to negotiate an exclusive strategic MOU and propose terms covering:

- exclusivity scope, including product family, territory (proposed: Nigeria), customer/channel segment and any reserved accounts or existing commitments;
- proposed term, renewal, performance milestones, review process and termination/transition arrangements;
- commercial framework, including pricing, discounts, minimum order commitments (if any), lead times, warranty, training, after-sales service, spare parts and local technical support;
- joint integration, localization, branding and marketing responsibilities, and ownership/licensing of software, API/SDK improvements, data and documentation;
- confidentiality, permitted disclosures, applicable law, dispute resolution and the status of provisions intended to be binding or non-binding.

Please identify which MOU provisions you propose to be binding and provide your standard MOU or draft terms for review. The scope and legal effect of exclusivity should be expressly stated in the signed agreement.

8 Requested response and language

Please provide all equipment and software supplied to us in English, including the software interface, menus, alerts and configuration screens. All user manuals, installation and configuration guides, API/SDK documentation, training materials, labels, technical correspondence and other documents supplied to us must also be in English. For each technical question, indicate “supported”, “not supported” or “requires configuration/custom development”, and name the relevant model, firmware/software version and any additional licence or hardware. Please propose a technical workshop to review the network topology and demonstrate the external interfaces.

We look forward to assessing the integration and discussing a long-term exclusive partnership.

Yours faithfully,